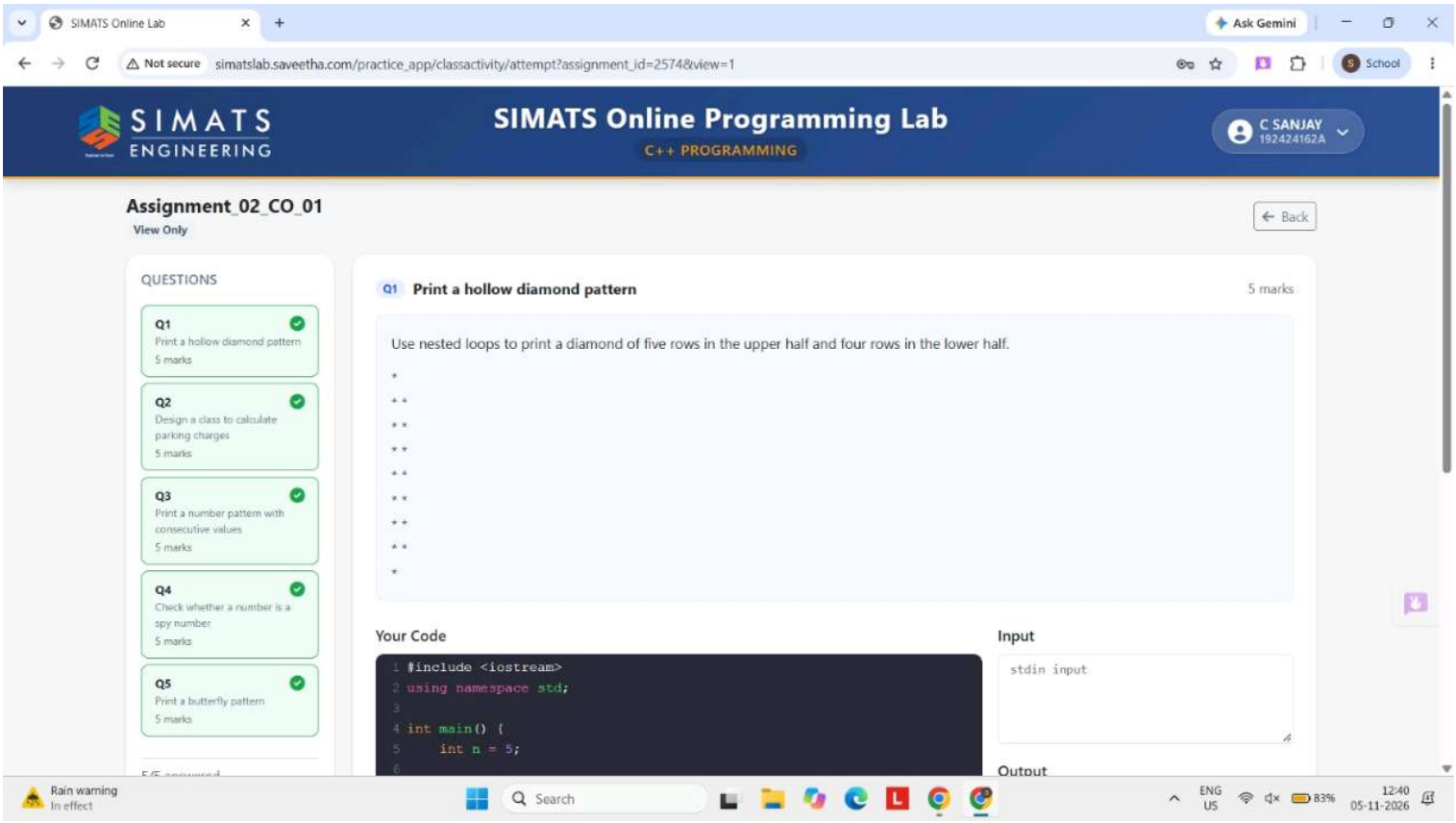

SIMATS Online Lab
Ask Gemini
Not secure
simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1
School
SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
C SANJAY
192424162A
Assignment_02_CO_01
View Only
Back
QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
Q1 Print a hollow diamond pattern
5 marks
Use nested loops to print a diamond of five rows in the upper half and four rows in the lower half.
*
* *
* *
* *
* *
* *
* *
*
Your Code
#include <iostream>
using namespace std;
int main() {
int n = 5;
Input
stdin input
Output
Rain warning
In effect
Search
ENG
US
83%
12:40
05-11-2026

SIMATS Online Lab

Ask Gemini

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1

School

SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

C SANJAY
192424162A

Check whether a number is a spy number
5 marks

Q5
Print a butterfly pattern
5 marks

5/5 answered

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 5;


    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "* ";
        }
        cout << endl;
    }


    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            cout << "* ";
        }
        cout << endl;
    }

    return 0;
}
```

Input

stdin input

Output

```
*
* *
* * *
* * * *
* * * * *
* * * *
```

Submitted

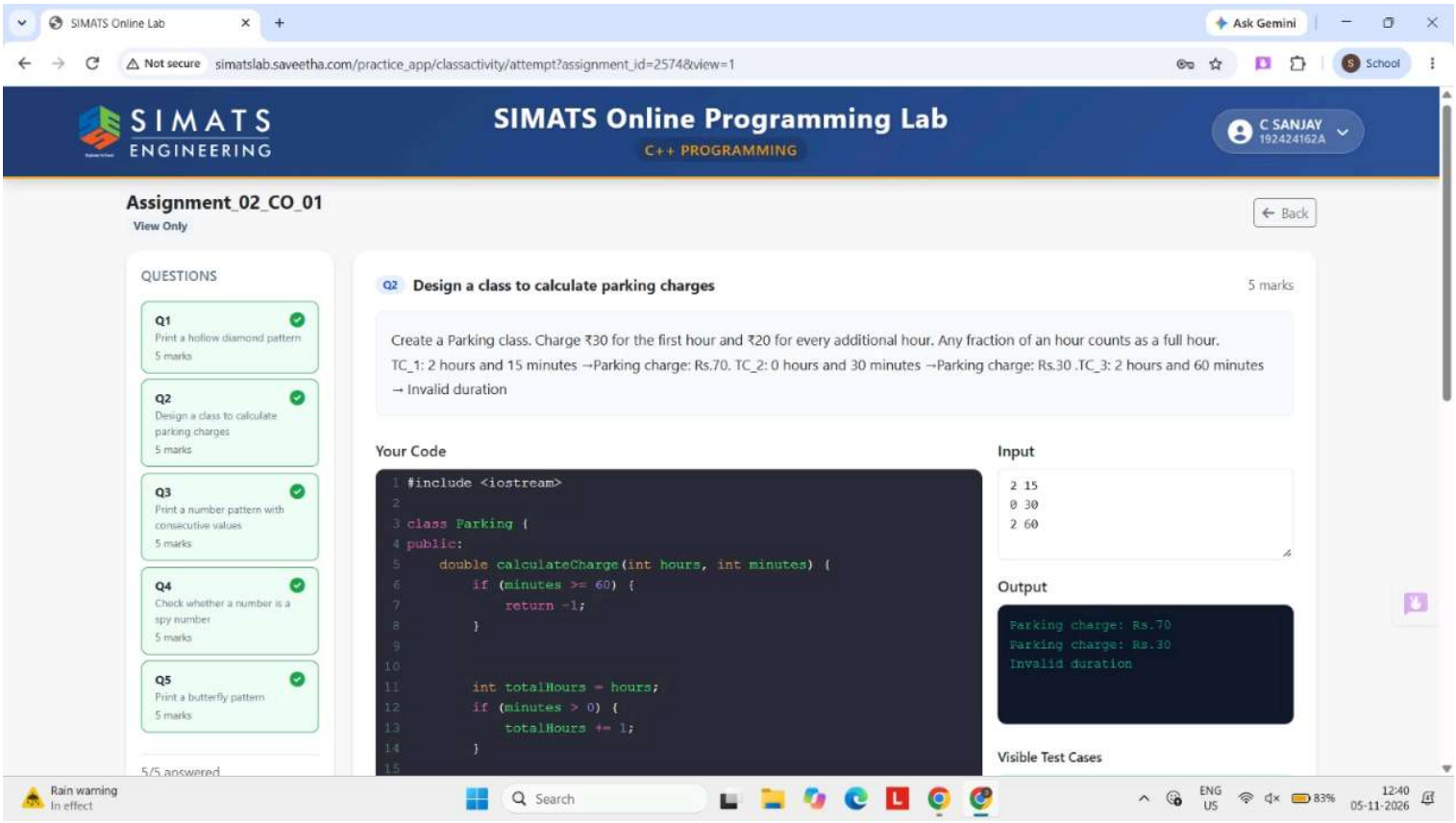
Rain warning
In effect

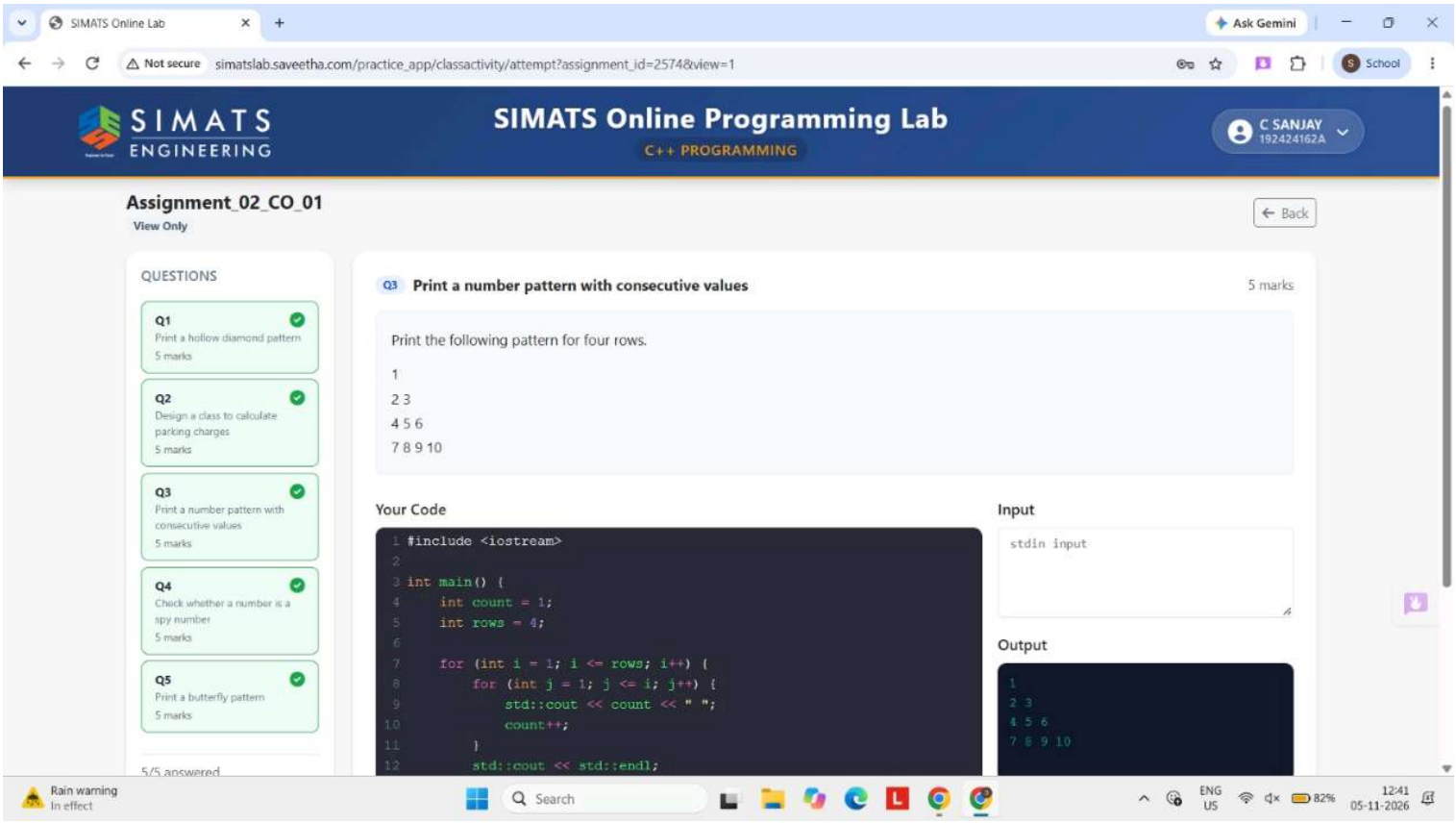
Search

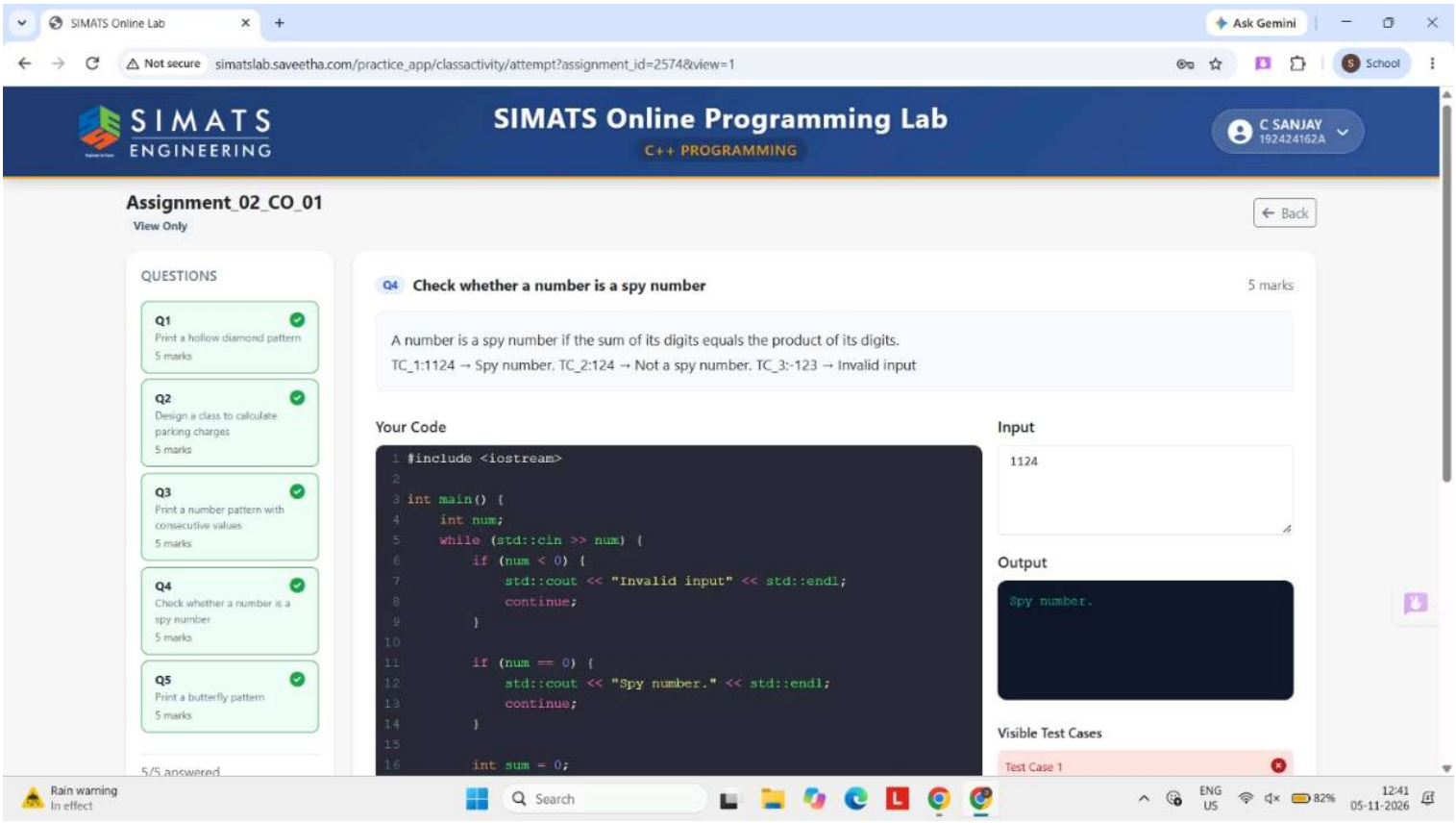
ENG
US

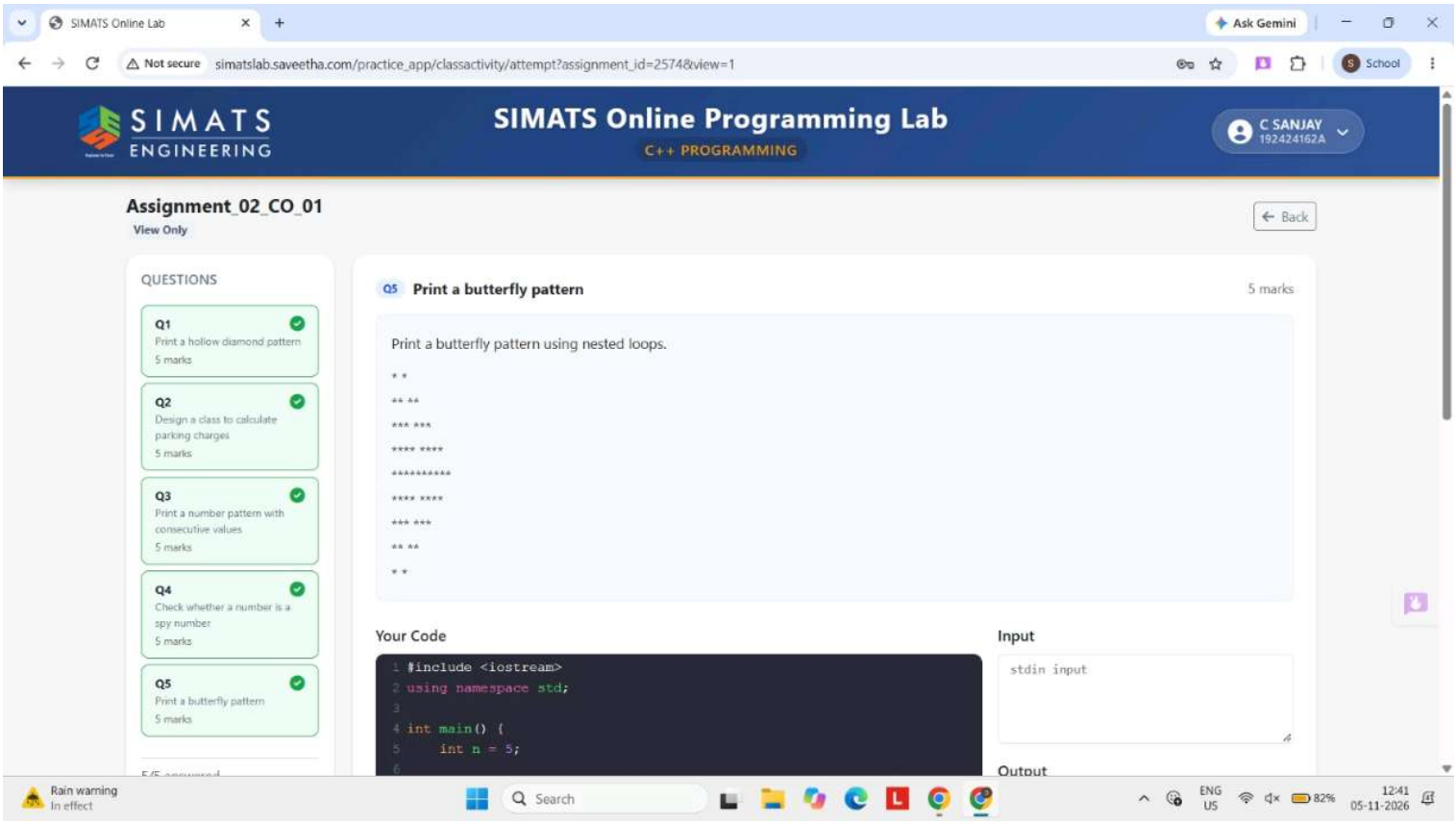
83%

12:40
05-11-2026


SIMATS Online Lab
Ask Gemini
Not secure
simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1
School
SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
C SANJAY
192424162A
Assignment_02_CO_01
View Only
Back
QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
5/5 answered
Q2 Design a class to calculate parking charges
5 marks
Create a Parking class. Charge ₹30 for the first hour and ₹20 for every additional hour. Any fraction of an hour counts as a full hour.
TC_1: 2 hours and 15 minutes →Parking charge: Rs.70. TC_2: 0 hours and 30 minutes →Parking charge: Rs.30 .TC_3: 2 hours and 60 minutes → Invalid duration
Your Code
#include <iostream>
class Parking {
public:
double calculateCharge(int hours, int minutes) {
if (minutes >= 60) {
return -1;
}
int totalHours = hours;
if (minutes > 0) {
totalHours += 1;
}
Input
2 15
0 30
2 60
Output
Parking charge: Rs.70
Parking charge: Rs.30
Invalid duration
Visible Test Cases
Rain warning
In effect
Search
ENG
US
83%
12:40
05-11-2026


SIMATS Online Lab
Ask Gemini
Not secure
simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1
School
SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
C SANJAY
192424162A
Assignment_02_CO_01
View Only
Back
QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
5/5 answered
Q3 Print a number pattern with consecutive values
5 marks
Print the following pattern for four rows.
1
2 3
4 5 6
7 8 9 10
Your Code
#include <iostream>
int main() {
int count = 1;
int rows = 4;
for (int i = 1; i <= rows; i++) {
for (int j = 1; j <= i; j++) {
std::cout << count << " ";
count++;
}
std::cout << std::endl;
Input
stdin input
Output
1
2 3
4 5 6
7 8 9 10
Rain warning
In effect
Search
ENG
US
82%
12:41
05-11-2026


SIMATS Online Lab
Ask Gemini
Not secure
simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1
School
SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
C SANJAY
192424162A
Assignment_02_CO_01
View Only
Back
QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
5/5 answered
Q4 Check whether a number is a spy number
5 marks
A number is a spy number if the sum of its digits equals the product of its digits.
TC_1:1124 → Spy number. TC_2:124 → Not a spy number. TC_3:-123 → Invalid input
Your Code
#include <iostream>
int main() {
int num;
while (std::cin >> num) {
if (num < 0) {
std::cout << "Invalid input" << std::endl;
continue;
}
if (num == 0) {
std::cout << "Spy number." << std::endl;
continue;
}
int sum = 0;
Input
1124
Output
Spy number.
Visible Test Cases
Test Case 1
Rain warning
In effect
Search
ENG
US
82%
12:41
05-11-2026


SIMATS Online Lab
Ask Gemini
Not secure
simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1
School
SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
C SANJAY
192424162A
Assignment_02_CO_01
View Only
Back
QUESTIONS
Q1
Print a hollow diamond pattern
5 marks
Q2
Design a class to calculate parking charges
5 marks
Q3
Print a number pattern with consecutive values
5 marks
Q4
Check whether a number is a spy number
5 marks
Q5
Print a butterfly pattern
5 marks
Q5 Print a butterfly pattern
5 marks
Print a butterfly pattern using nested loops.
* *
** **
*** ***
**** ****
****
**** ****
*** ***
** **
* *
Your Code
#include <iostream>
using namespace std;
int main() {
int n = 5;
Input
stdin input
Output
Rain warning
In effect
Search
ENG
US
82%
12:41
05-11-2026

SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=2574&view=1

SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

C SANJAY 192424162A

Q4
Check whether a number is a spy number
5 marks

Q5
Print a butterfly pattern
5 marks

5/5 answered

* *

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 5;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        for (int j = 1; j <= 2 * (n - i); j++) {
            cout << " ";
        }
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }

    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        for (int j = 1; j <= 2 * (n - i); j++) {
```

Input

stdin input

Output

```
*         *
**       **
***     ***
****   ****
**********
****   ****
```

Rain warning In effect

Search

ENG US 82% 12:42 05-11-2026